Vision-Language Model: What & Why

Intelligent OCR for scanned documents (Francais - English - Arabic) | Technical rationale for the OCR engine selection

A Vision-Language Model (VLM) reads an image and produces text in a single neural network - unlike classical OCR (e.g. Tesseract), which detects character shapes with no understanding of context, layout, or language. That understanding is what lets it handle messy scans, mixed scripts, tables, and handwriting.

What we actually use

Role	Model	Where it runs
Default engine	Qwen3-VL 4B (instruct) - qwen3-vl:4b-instruct	Local, via Ollama, on the GPU
Second opinion (QA)	Qwen3-VL 8B (instruct) - qwen3-vl:8b-instruct	Local, via Ollama
Optional cloud	Llama 4 Scout (vision)	Groq API - currently unavailable

Why Qwen3-VL (the model family)

Strong multilingual + Arabic: The project targets French, English, and Arabic (right-to-left). Qwen models are trained heavily on multilingual and Arabic data and are among the best open VLMs for non-Latin scripts - confirmed on the mixed FR/AR documents.

Built for document/OCR tasks: Handles dense text, tables, and structured forms, and outputs Markdown structure - exactly what the safeguarding form (tables, checkboxes) needed.

Open weights, runs locally: Usable through Ollama with no license fee.

Why the 4B, and why the instruct variant

4B fits the hardware: The laptop has an NVIDIA RTX 4050 with 6 GB VRAM. The 4B model (~3.3 GB) stays fully resident on the GPU; larger models spill to CPU and slow down.

Measured, not assumed: On the dense handwritten form: 4B scored 10/10 completeness in 59 s; 8B scored 9/10 in 135 s (it spilled to CPU on 6 GB). So 4B is faster AND more complete on this machine.

The -instruct tag is mandatory: Base qwen3-vl tags run in 'thinking' mode - they spend the whole token budget on reasoning and return an empty transcription. The instruct variant transcribes directly.

Why local (Ollama) is the default engine

Confidentiality: The test document is a strictly confidential safeguarding form with names and medical details. Local means no document ever leaves the machine - the decisive factor for data governance.

No cost, no limits, offline: No per-page API fees, no rate limits, works with no internet.

Fast enough on real hardware: ~30 s per page after tuning the context window (num_ctx=8192 tripled speed versus the naive setting).

Why Groq (cloud) is kept as an option - and what happened

Benchmark / fallback: A much larger model (Llama 4 Scout, 17B) for cases that trade confidentiality for maximum accuracy.

Reality check: As of mid-2026, Groq removed all Llama-4 vision models from general access, so that engine currently 404s. This reinforces the local-first choice: the local engine needs nothing external.

Summary

The system uses Qwen3-VL-4B-Instruct, an open vision-language model served locally via Ollama, as its default OCR engine. It was chosen for its strong multilingual and Arabic performance, its document/table/handwriting handling, and - critically - because it runs fully offline on a 6 GB laptop GPU, keeping confidential documents on-device. The 4B size was selected over the 8B after direct measurement showed it was faster and more complete on the target hardware. A cloud engine (Groq / Llama-4) is retained as an optional higher-capacity fallback, though vision access was withdrawn by the provider, underscoring the value of the local-first design.